

# Model review and revisions

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I reviewed the model and calibration, including preferences, credit, demographic accounting, and the empirical targets.

- Returned to a Stone–Geary housing requirement that applies only to parents, retaining nonlinear benefits from children.
- Tested equivalence scales and changing housing shares.
- Rebuilt the first-birth housing evidence and national input measures.
- Revised adult entry and tested how purchase timing affects the calibration.

# Household decisions

Overlapping generations of households differ in age, earnings, wealth, housing tenure, and family size.

- During fertile ages, households choose whether to attempt a birth; success becomes less likely with age.
- They choose consumption, saving, housing size, and whether to rent or own; renters pay rent, while owners make a down payment and face selling costs.
- Persistent earnings risk and borrowing limits shape saving; a first-birth fixed cost and taste shocks shape fertility.
- Households retire, face mortality risk, and value bequests.

## Children and housing needs

The first child requires basic space at home; each additional child raises the goods needs of a larger family.

$$u(c, s; m) = \frac{\left[ \frac{c^{\alpha_0} s^{1-\alpha_0}}{e(m)} \right]^{1-\sigma}}{1-\sigma} + \psi m^{1-\gamma},$$
$$e(m) = \left( \frac{2 + 0.7m}{2} \right)^{0.7}, \quad s = \eta_\tau [h - h_P \mathbf{1}\{m > 0\}],$$
$$\eta_R = 1, \quad \eta_O = \chi.$$

Here  $h$  is rooms,  $m$  is children at home, and  $\tau \in \{R, O\}$  denotes tenure. Parents require  $h > h_P$ ;  $\chi$  raises housing services for owners. The equivalence scale  $e(m)$  increases family consumption needs. Child benefits are zero at  $m = 0$ ;  $\psi$  sets their level and  $\gamma$  controls diminishing marginal benefits.

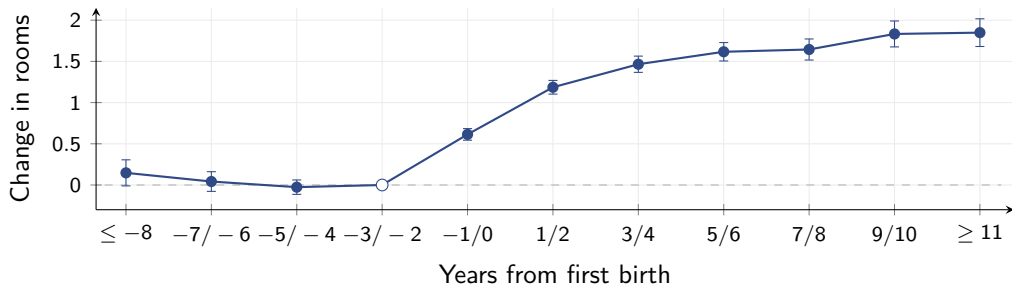
# Demographic and market equilibrium

Children leaving home and new adult households entering the economy are separate transitions.

- Children leave dependency independently; adult entry follows earlier births with a lag.
- Positive estates finance entrant assets; any remainder leaves the economy.
- Stationary calibration normalizes population to one: the house price satisfies birth renewal, and housing clearing determines  $H_0$ .
- A balanced payroll tax finances equal retiree pensions.

## Housing around the first birth

We found a mistake in the earlier regression and rebuilt the PSID event study using interview dates and household status before birth.



At years 3–4, rooms rise by **1.465** (SE 0.050), ownership by **26.886 pp**, and recent moving falls by **21.147 pp**.

Reference person or spouse before birth. Baseline: years  $-3/-2$ ; person and year effects; survey weights; 95% pointwise intervals.

## Credit timing

A renter with financial wealth  $b$  and period income  $y$  who buys a home worth  $Q = pH$ :

$$c + (\delta + \tau_H) Q + b' = R(b - Q) + y.$$

Normally the buyer must have the 20% down payment at closing:  $b \geq (1 - \phi)Q$ , with  $\phi = 0.8$ .

**The problem.** A model period is four years. Requiring the down payment from wealth at the start of the period ignores saving within the period; counting all of it lets a buyer close with little equity and reach 20% only over the next four years. The rules differ in how much of the period's net saving  $S = b' - (b - Q)$  counts:

$$b + \lambda S \geq (1 - \phi) Q.$$

- Soft,  $\lambda = 1$ : all of the period's saving counts.
- Quarter-saving,  $\lambda = \frac{1}{4}$ : a quarter of it counts, roughly one year of saving.
- Hard,  $\lambda = 0$ : only wealth brought into the period counts.

## Externally set parameters

| Parameter                            | Value                | Source                         |
|--------------------------------------|----------------------|--------------------------------|
| Model period                         | 4 years              |                                |
| Entry and retirement ages            | 18, 66               |                                |
| Risk aversion $\sigma$               | 2                    |                                |
| Equivalence scale $(\omega, \nu)$    | (0.7, 0.7)           | Scholz et al. (2006)           |
| Consumption share $\alpha_0$         | 0.733                | CEX 2019–23, childless renters |
| Annual real interest rate            | 2%                   | Greaney et al. (2025)          |
| Annual depreciation; property tax    | 1.416%; 1.060%       | National housing inputs        |
| Financed share $\phi$                | 0.80                 | 20% down payment               |
| Selling cost $\psi_s$                | 6%                   | Greaney et al. (2025)          |
| Owner house sizes                    | 2, 4, 6, 8, 10 rooms |                                |
| Largest rental unit                  | 6 rooms              | Greaney et al. (2025)          |
| Housing-supply elasticity            | 0.63                 | Baum-Snow and Han (2024)       |
| Pension / working-household earnings | 0.229                | CPS ASEC, income 2007          |
| Payroll tax                          | 8.0%                 | Balances the pension budget    |
| Bequest wealth shift $\theta_1$      | 0.008                | 1% of median annual earnings   |

# Parameters and target moments

2007 stationary economy: ten parameters, ten moments.

| Parameter                  | Interpretation               | Main target moment                        | Target |
|----------------------------|------------------------------|-------------------------------------------|--------|
| <i>Saving and bequests</i> |                              |                                           |        |
| $\beta$                    | Patience                     | Wealth / earnings                         | 6.927  |
| $\theta_0$                 | Bequest weight               | Annual bequests / wealth (%) <sup>*</sup> | 0.729  |
| <i>Housing and tenure</i>  |                              |                                           |        |
| $\chi$                     | Owner housing premium        | Ownership, ages 30–55 (%)                 | 67.626 |
| $\kappa_T$                 | Tenure tastes                | Recent-parent ownership difference (pp)   | 12.761 |
| $h_P$                      | Basic space for parents      | First-birth rooms difference <sup>*</sup> | 1.465  |
| $\chi, h_P$                | Jointly                      | Mean rooms <sup>*</sup>                   | 5.729  |
| <i>Fertility</i>           |                              |                                           |        |
| $\psi$                     | Child-benefit level          | Childlessness, ages 40–44 (%)             | 19.828 |
| $\xi$                      | First-birth cost             | Children at age 25                        | 0.810  |
| $\kappa_1$                 | First-birth taste dispersion | Mean first-birth age                      | 25.976 |
| $\kappa_C$                 | Later-birth taste dispersion | One child among mothers (%)               | 21.366 |
| $\gamma$                   | Child-benefit curvature      | No distinct moment                        |        |

All ten are estimated jointly. Housing has four moments and three parameters; fertility has four and five.

<sup>\*</sup>Recomputed: mean rooms (AHS), bequests (SCF), first-birth rooms (PSID).

# Stone–Geary calibration

2007 stationary economy; 80% mortgage financing

| Moment                                      | Target       | Soft         | Hard         | Quarter-saving |
|---------------------------------------------|--------------|--------------|--------------|----------------|
| Completed fertility                         | 2.100        | 2.100        | 2.100        | 2.100          |
| Childlessness, ages 40–44 (%)               | 19.828       | 20.222       | 20.413       | 20.546         |
| One child among mothers, ages 40–44 (%)     | 21.366       | 21.264       | 20.907       | 20.844         |
| Mean first-birth age (years)                | 25.976       | 26.032       | 26.007       | 26.057         |
| Wealth / earnings                           | 6.927        | 6.551        | 6.610        | 6.645          |
| Annual child bequests / wealth (%)          | 0.729        | 0.685        | 0.684        | 0.685          |
| <b>Mean occupied rooms</b>                  | <b>5.729</b> | <b>5.957</b> | <b>6.241</b> | <b>6.131</b>   |
| Ownership, ages 30–55 (%)                   | 67.626       | 66.527       | 60.878       | 64.572         |
| <b>First-birth rooms difference</b>         | <b>1.465</b> | <b>1.324</b> | <b>1.063</b> | <b>1.167</b>   |
| Recent-parent ownership difference (pp)     | 12.761       | 11.825       | 10.125       | 11.683         |
| <b>Children at age 25 (capped at three)</b> | <b>0.810</b> | <b>0.528</b> | <b>0.524</b> | <b>0.521</b>   |
| Validation moments                          | Target       | Soft         | Hard         | Quarter-saving |
| First births at age 30+ (%)                 | 24.928       | 23.448       | 23.078       | 23.395         |
| Older wealth/income p90/p50                 | 3.516        | 2.968        | 2.861        | 2.937          |
| Family rooms difference                     | 0.385        | 0.304        | 0.321        | 0.316          |

## Parameter estimates

Estimates under the three purchase rules;  $H_0$  follows from housing clearing.

| Parameter  | Economic interpretation                     | Soft  | Hard  | Quarter-saving |
|------------|---------------------------------------------|-------|-------|----------------|
| $\beta$    | Annual discount factor                      | 0.967 | 0.967 | 0.968          |
| $\chi$     | Owner housing-service premium               | 1.097 | 1.068 | 1.091          |
| $\xi$      | First-birth fixed utility cost              | 0.352 | 0.385 | 0.397          |
| $\kappa_1$ | First-birth taste dispersion                | 0.124 | 0.123 | 0.128          |
| $\kappa_C$ | Later-birth taste dispersion                | 0.363 | 0.350 | 0.348          |
| $\theta_0$ | Bequest weight                              | 0.106 | 0.106 | 0.109          |
| $\gamma$   | Child-benefit curvature                     | 0.101 | 0.101 | 0.098          |
| $\kappa_T$ | Tenure-choice dispersion                    | 0.014 | 0.014 | 0.015          |
| $\psi$     | Child-benefit level                         | 0.178 | 0.172 | 0.174          |
| $h_P$      | Basic space requirement for parents (rooms) | 2.504 | 2.562 | 2.492          |
| $H_0$      | Housing supply coefficient                  | 6.757 | 7.546 | 7.289          |

## More credit raises ownership but not births

Quarter-saving rule; financed share 80% vs. 100%; house price and parameters held fixed

|                                                                    | 80% financing | 100% financing |
|--------------------------------------------------------------------|---------------|----------------|
| Ownership, ages 30–55 (%)                                          | 64.6          | 74.0           |
| First-birth flow (relative change)                                 |               | –0.5%          |
| Completed fertility                                                | 2.100         | 2.090          |
| <i>Households entering at ages 18–21 (same households in both)</i> |               |                |
| Ownership (%)                                                      | 10.5          | 31.4           |
| First-birth probability (%)                                        | 25.5          | 25.8           |

Recalibrating under the hard and quarter-saving rules gives the same sign: first births fall by about 0.5%.

## Why the response is small: what we have checked

- Rent per room equals the owner's cost per room, and  $\chi$  does not depend on children. Credit makes buying easier but does not make a child's space cheaper.
- 60% of renting parents are at the six-room cap. Without the cap the median would rent 7.3 rooms. Holding saving fixed, the cap costs them about 2% of consumption in that period.
- The parent requirement  $h_P$  is the same for one, two or three children.

These are candidate explanations. We have not yet shown how much of the small response each one accounts for.

## What I want to try next

- Selling costs: check whether the cost of trading up after a birth is what keeps credit from reaching fertility, and whether the 6% seller-paid cost is the right convention.
- Small owner homes: young buyers may lock into homes too small for a family.
- A housing requirement that grows with the number of children.
- Recalibrate before drawing conclusions from any of these.